Data Science

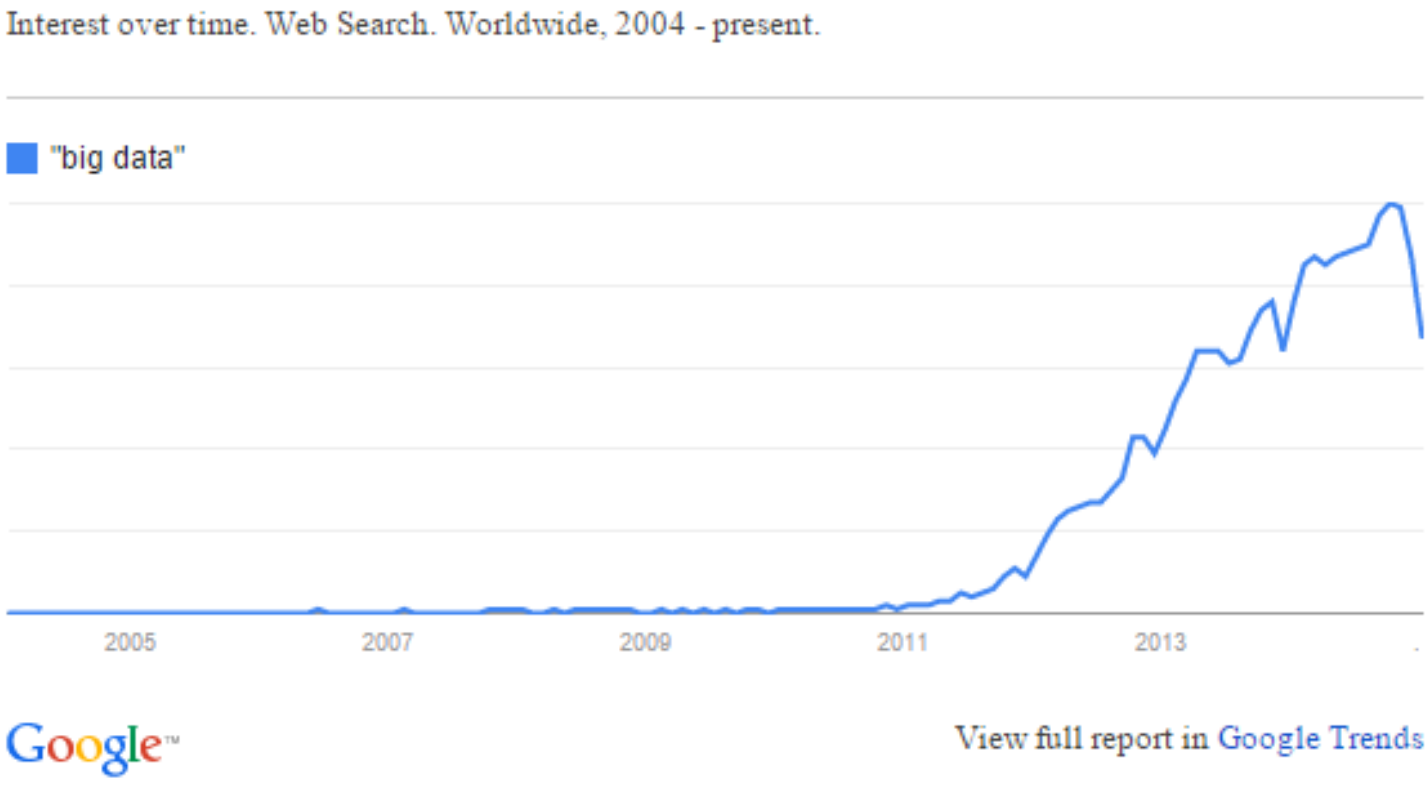
Gaurav Sood

Big Data

Big Data

Big data attracted widespread attention in the 2010s.

Big Data


Interest over time. Web Search. Worldwide, 2004 - present.
"big data"
2005
2007
2009
2011
2013
Google™
View full report in Google Trends

Big Data

But where's the cheese?

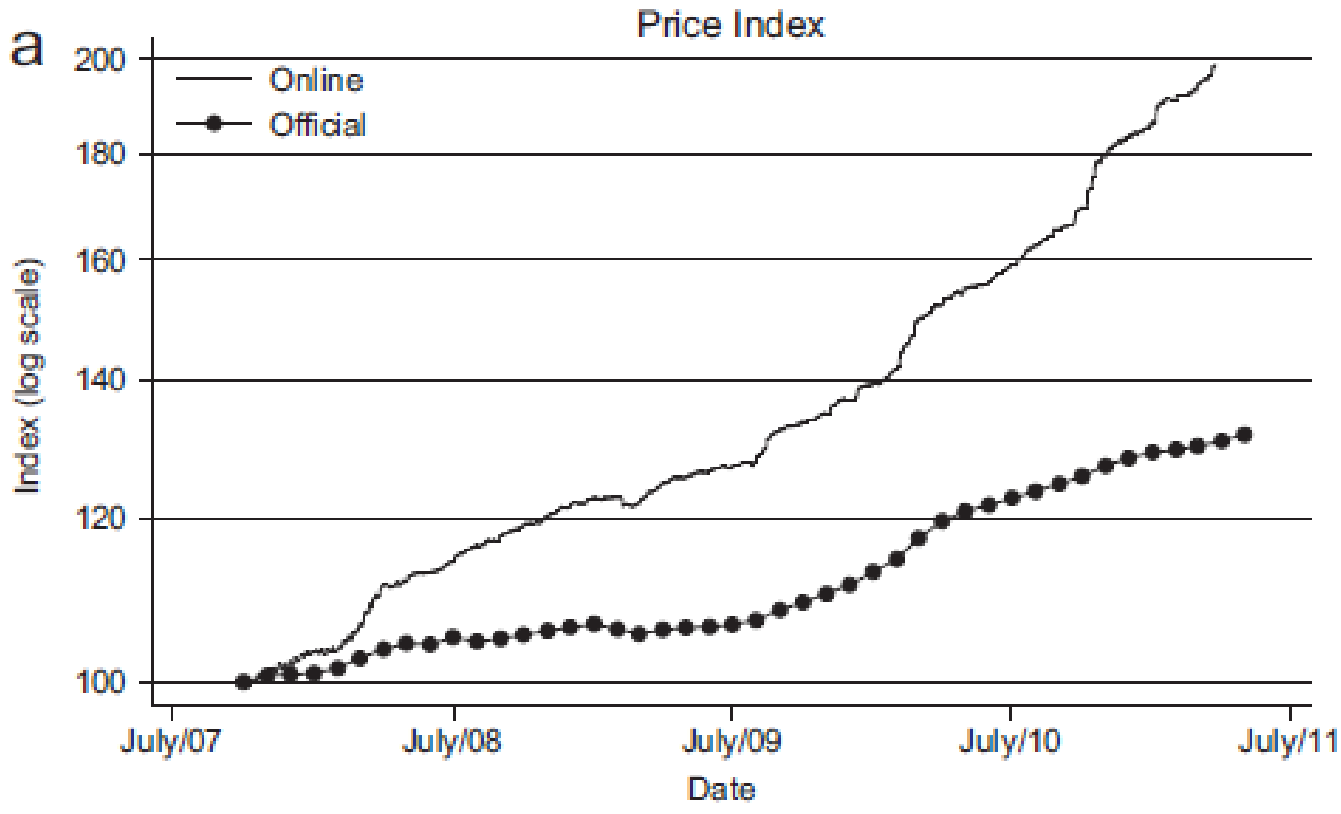
Some examples

Fishing Out Fishy Figures



"The CPINu's August inflation figure of 1.3% is less than half the 2.65% of the CPI Congreso, a compilation of private estimates gathered by opposition members of Congress." (Economist)

Fishing Out Fishy Figures



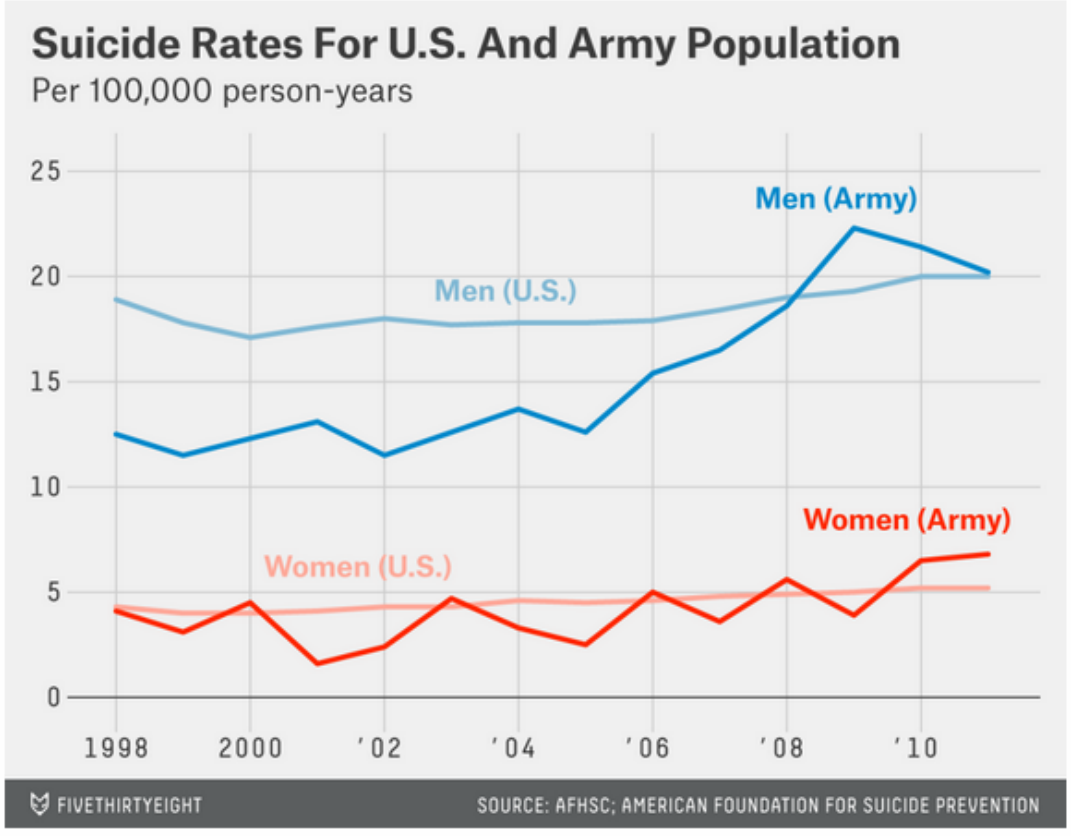
Source: Online vs Official Price Indexes: Measuring Argentina's Inflation By Alberto Cavallo

Suicide Risk in Army STARRS

Suicide Risk in Army STARRS

Can administrative data help identify elevated suicide risk after psychiatric hospitalization?

Suicide Risk in Army STARRS



Suicide Risk in Army STARRS

STARRS studied U.S. Army soldiers after psychiatric hospital discharge, not all soldiers.

Suicide Risk in Army STARRS

The highest-risk 5% of hospitalizations accounted for 52.9% of suicides within 12 months of discharge. Predictive concentration alone does not establish that an intervention prevents suicide.

538 Article
STARRS paper

Predictive Policing: Historical Examples

Minority Report

Predictive Policing: Historical Examples

Predictive, ‘CompStat’, ‘HotSpot’ Policing

Predictive Policing: Historical Examples

PredPol: Predictive Policing

LAPD, Atlanta PD

Based off earthquake prediction algorithm

Predictive Policing: Historical Examples

"During a four-month trial in Kent, 8.5% of all street crime occurred within PredPol's pink boxes, with plenty more next door to them; predictions from police analysts scored only 5%. An earlier trial in Los Angeles saw the machine score 6% compared with human analysts' 3%."

Economist

Predictive Policing: Historical Examples



Web Search

Web Search


YAHOO!
What's New
Check Email
Personalize
Help
Yahoo! Pager
instant messaging
Yahoo! Pager
now works with chat
Yahoo! Mail
free email for life
Search
advanced search
Yahoo! Auctions - 1000's of items to bid on - Pokemon, Beanie Babies, video games, Furbys...
Shopping - Yellow Pages - People Search - Maps - Travel Agent - Classifieds - Personals - Games - Chat
Email - Calendar - Pager - My Yahoo! - Today's News - Sports - Weather - TV - Stock Quotes - more...
Arts & Humanities
Literature, Photography...
News & Media
Full Coverage, Newspapers, TV...
Business & Economy
Companies, Finance, Jobs...
Recreation & Sports
Sports, Travel, Autos, Outdoors...
Computers & Internet
Internet, WWW, Software, Games...
Reference
Libraries, Dictionaries, Quotations...
Education
College and University, K-12...
Regional
Countries, Regions, US States...
Entertainment
Cool Links, Movies, Humor, Music...
Science
Biology, Astronomy, Engineering...
Government
Military, Politics, Law, Taxes...
Social Science
Archaeology, Economics, Languages...
Health
Medicine, Diseases, Drugs, Fitness...
Society & Culture
People, Environment, Religion...
In the News
NATO - Serbia war
Giant bacterium discovered
Lakers release Rodman
more...
Marketplace
Charity Auctions - for the Kosovo relief effort
Find a new job!
more...
Inside Yahoo!
Y! Movies - showtimes, reviews
Y! Clubs - create your own
Y! Visa - instant credit while you wait
more...

Web Search

– Human Curation, Ad-hoc automation

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- Page Rank

Side-effects of Drugs

Side-effects of Drugs

"Adverse drug events cause substantial morbidity and mortality and are often discovered after a drug comes to market."

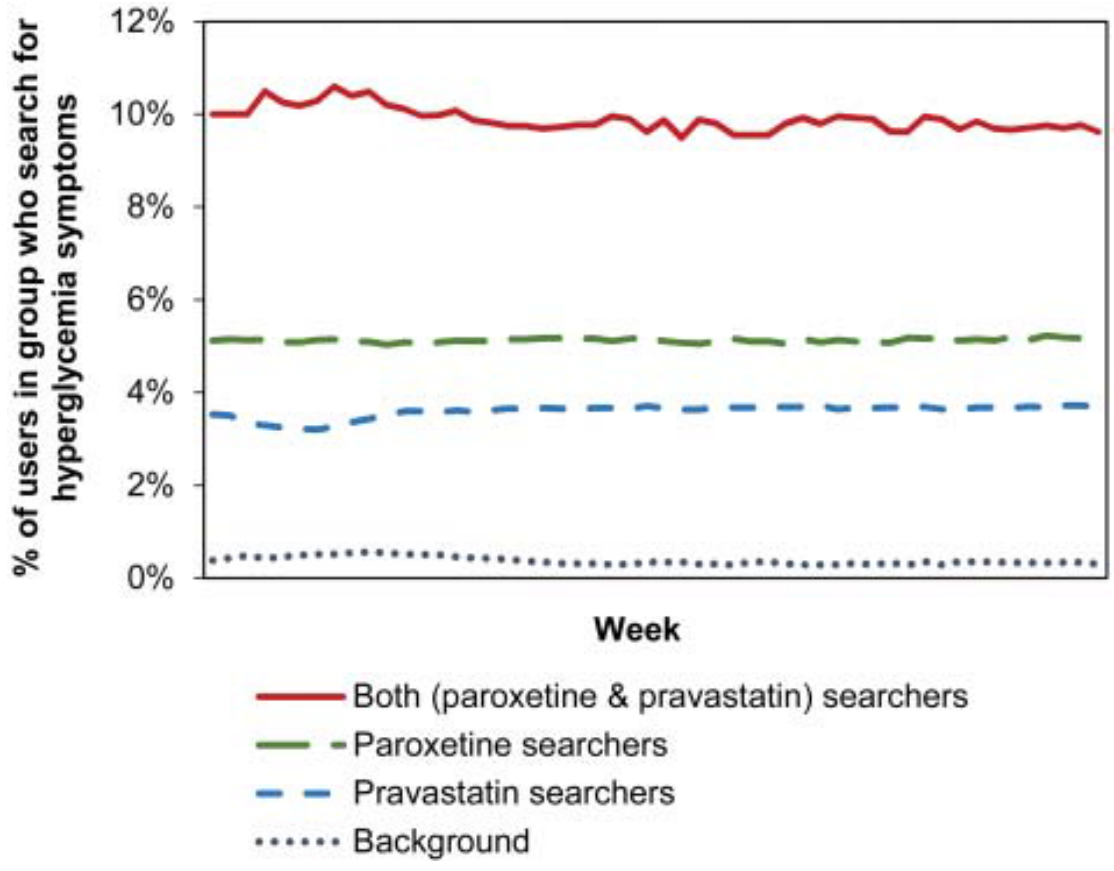
Side-effects of Drugs

FDA collects this information from “physicians, pharmacists, patients, and drug companies” but these reports “are incomplete and biased”

Side-effects of Drugs

"paroxetine and pravastatin, whose interaction was reported to cause hyperglycemia after the time period of the online logs used in the analysis"

Side-effects of Drugs



Web-scale Pharmacovigilance: Listening to Signals from the Crowd. By White et al.

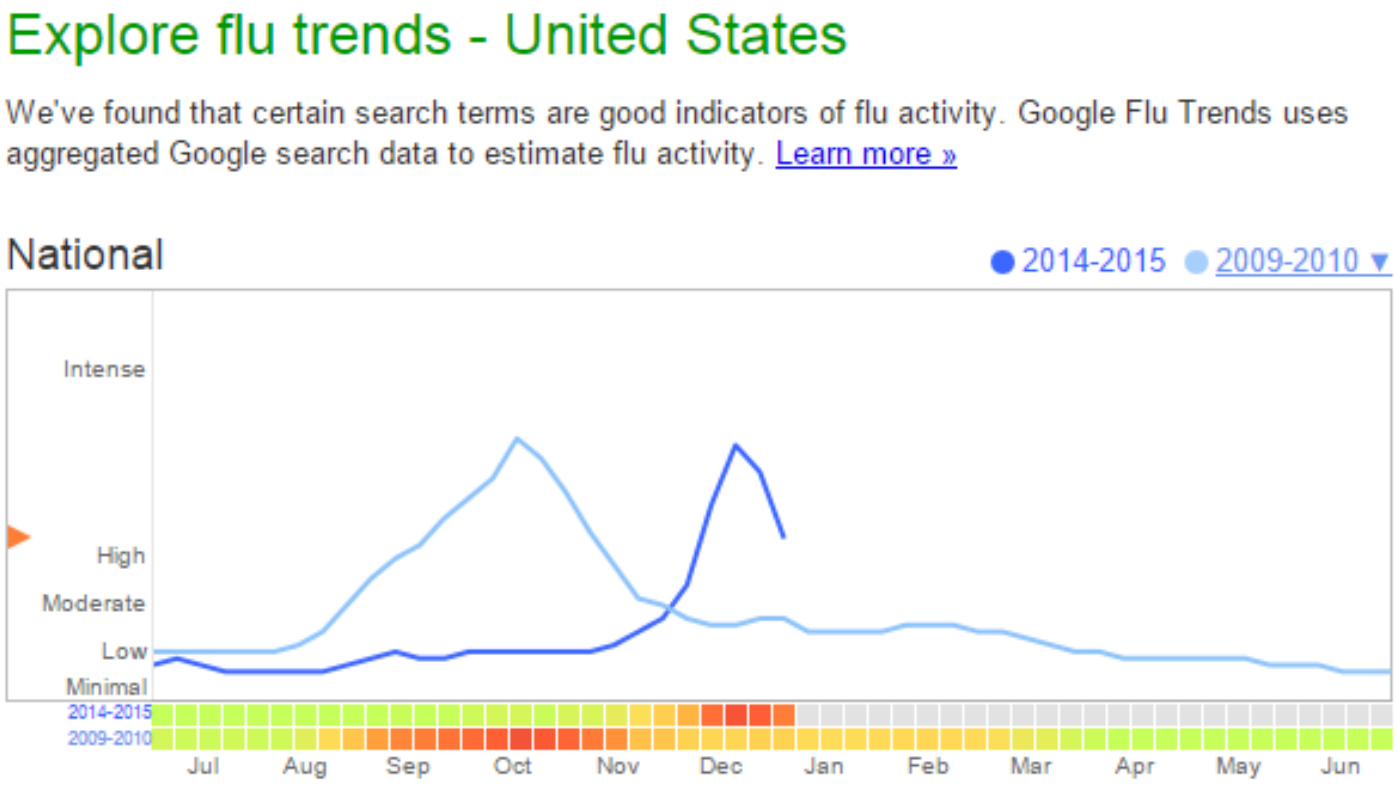
Flu Season

How many got the sniffles?

Flu Season

How many got the sniffles in the past month?

Flu Season

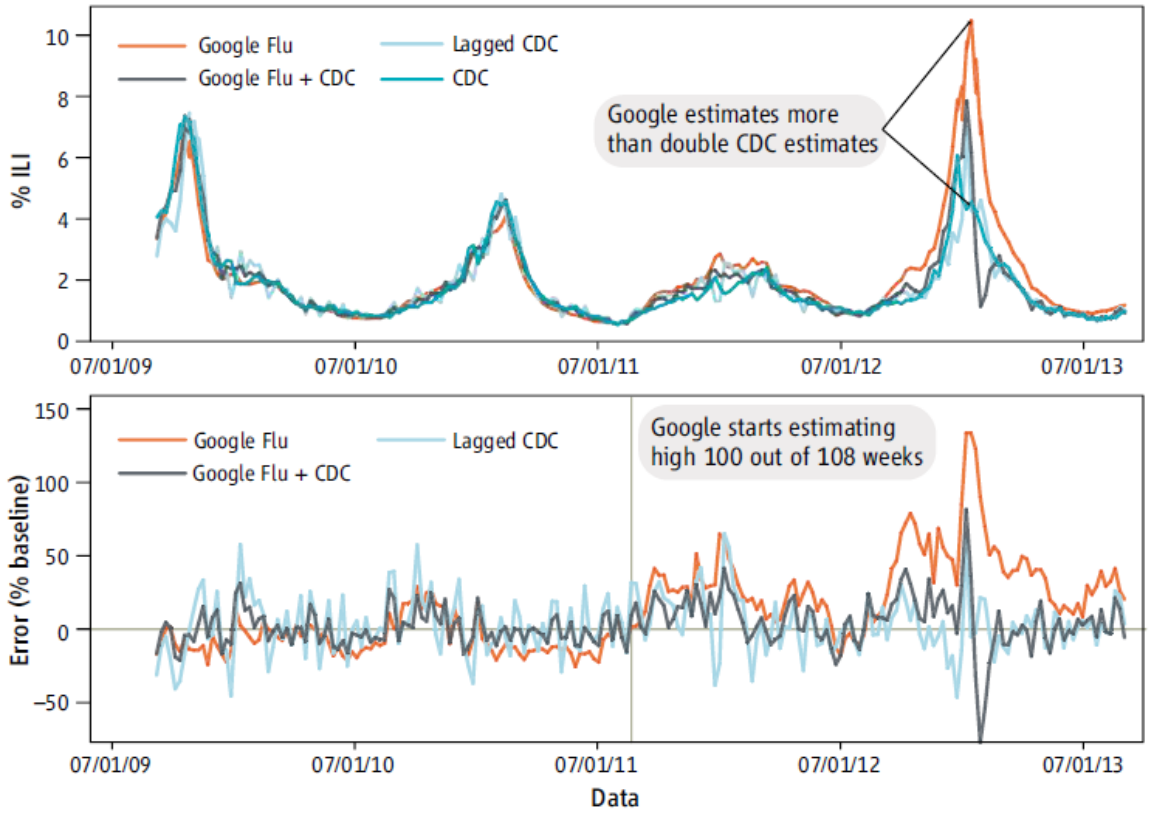


Google Flu Trends

Flu Season

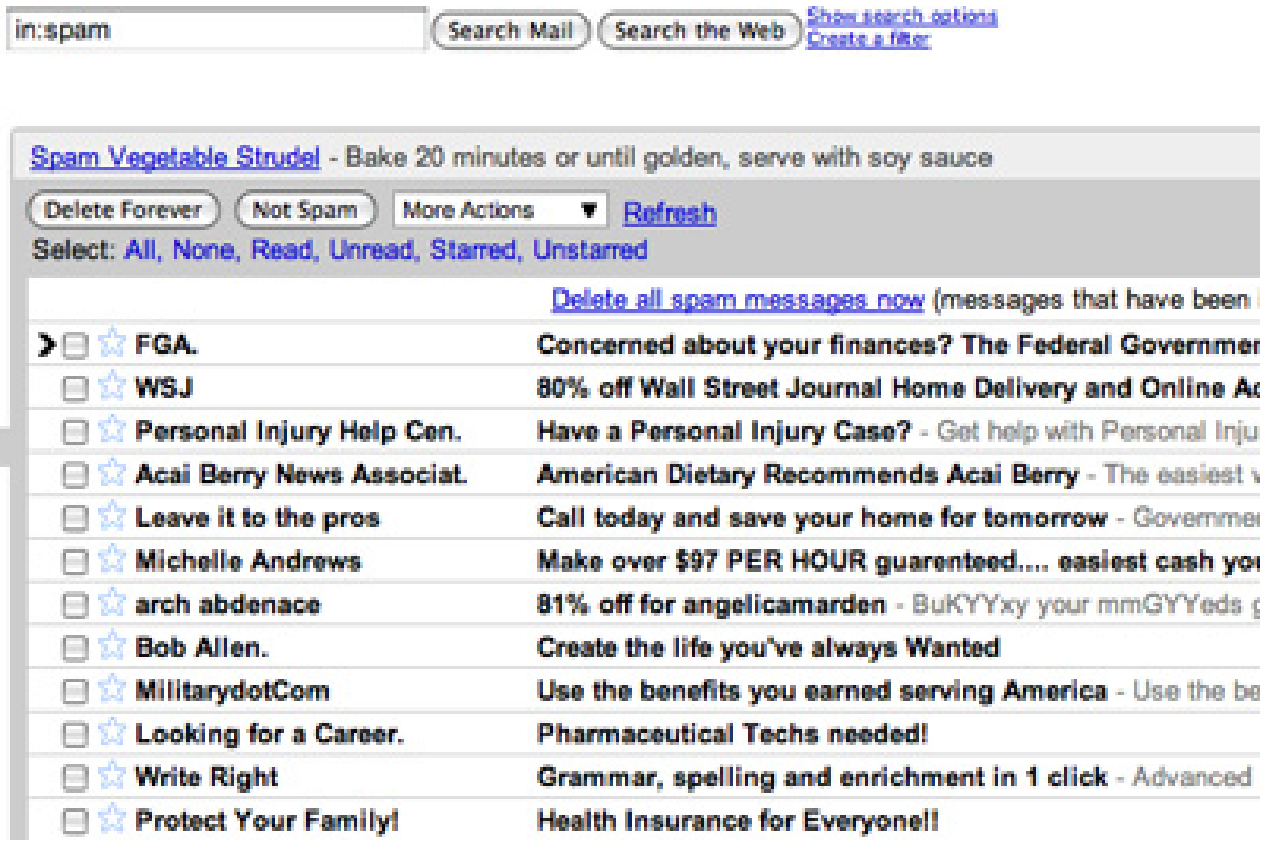
Google Flu is sick.

Flu Season



The Parable of Google Flu: Traps in Big Data Analysis. By Lazer et al.

Spam or Ham


in:spam
Search Mail
Search the Web
Show search options
Create a filter
Spam Vegetable Strudel - Bake 20 minutes or until golden, serve with soy sauce
Delete Forever
Not Spam
More Actions
Refresh
Select: All, None, Read, Unread, Starred, Unstarred
Delete all spam messages now (messages that have been
FGA. Concerned about your finances? The Federal Governmer
WSJ 80% off Wall Street Journal Home Delivery and Online Ac
Personal Injury Help Cen. Have a Personal Injury Case? - Get help with Personal Inju
Acai Berry News Associat. American Dietary Recommends Acai Berry - The easiest
Leave it to the pros Call today and save your home for tomorrow - Governme
Michelle Andrews Make over $97 PER HOUR guarenteed.... easiest cash yo
arch abdenace 81% off for angelicamarden - BuKYYxy your mmGYYeds
Bob Allen. Create the life you've always Wanted
MilitarydotCom Use the benefits you earned serving America - Use the be
Looking for a Career. Pharmaceutical Techs needed!
Write Right Grammar, spelling and enrichment in 1 click - Advanced
Protect Your Family! Health Insurance for Everyone!!

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- "According to Commtouch's Internet Threats Trend Report for the first quarter of 2013, an average of 97.4 billion spam e-mails and 973 million malware e-mails were sent worldwide each day in Q1 2013 (h/t Softpedia)."

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- $logit[p(spam)] = \alpha + f'\beta$ where f is frequencies.

Vote

A 61-million-person experiment in social influence and political mobilization

Robert M. Bond[1], Christopher J. Fariss[1], Jason J. Jones[2], Adam D. I. Kramer[3], Cameron Marlow[3], Jaime E. Settle[1] & James H. Fowler[1,4]

Human behaviour is thought to spread through face-to-face social networks, but it is difficult to identify social influence effects in observational studies[9–13], and it is unknown whether online social

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The social message increased validated turnout by 0.39 percentage points relative to the informational message (Bond et al., 2012).

Vote for Obama

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- Used SQL database Vertica for 'speed-of-thought' analyses.

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- 'Big data' is high **volume**, high **velocity** and high **variety** information assets that demand cost-effective, innovative forms of information processing[.]
  Gartner, Inc.'s "3Vs" definition.

Sources of (Big) Data

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- Data as the primary goal of activities

  Telescopes, Genetic sequencers, 61 million person experiments . . .

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Implications for Statistics, Computation.

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  Causal claims still require a credible design and assumptions

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- Ideal parallel speedup divides work by m computers; communication and serial work limit the gain.
- For really big data, complexity matters
  Tree traversal: $O(N)$; balanced lookup: $O(\log N)$; comparison sorting: $O(N \log N)$

  Streaming application

MapReduce and PageRank

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  - Distributed programming can be very very hard.

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  - Send different words to different computers
  - Combine

```
  map(String input_key, String input_value):
  // input_key: document name
  // input_value: document contents
  for each word w in input_value:
    EmitIntermediate(w, "1");


reduce(String output_key, Iterator intermediate_values):
  // output_key: a word
  // output_values: a list of counts
  int result = 0;
  for each v in intermediate_values:
    result += ParseInt(v);
  Emit(AsString(result));
```

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  - Web - lots of false positives. Some of it malware peddling sites.

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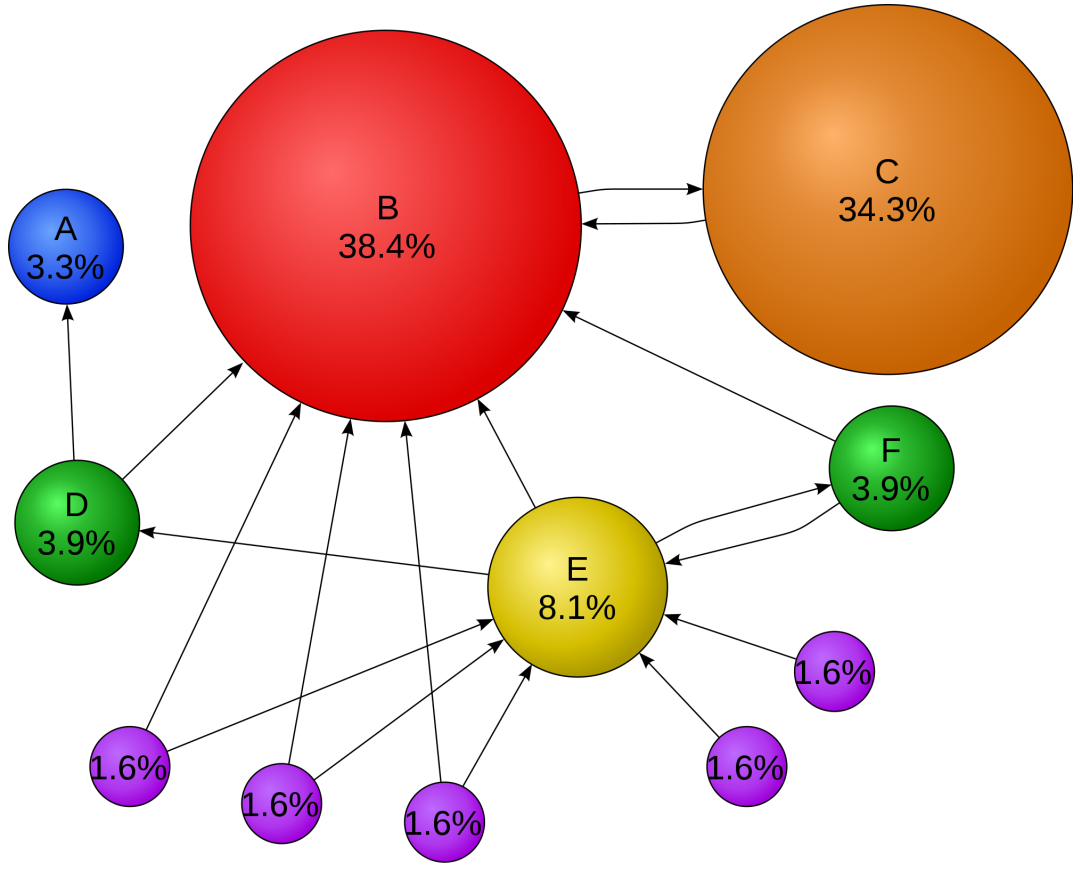
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  - Say page i has importance r_i, and n outgoing links, each vote $= r_i/n$

Page Rank Example



Source: Wikipedia

PageRank

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over j where j tracks all the pages pointing to i.

PageRank: Random Surfer

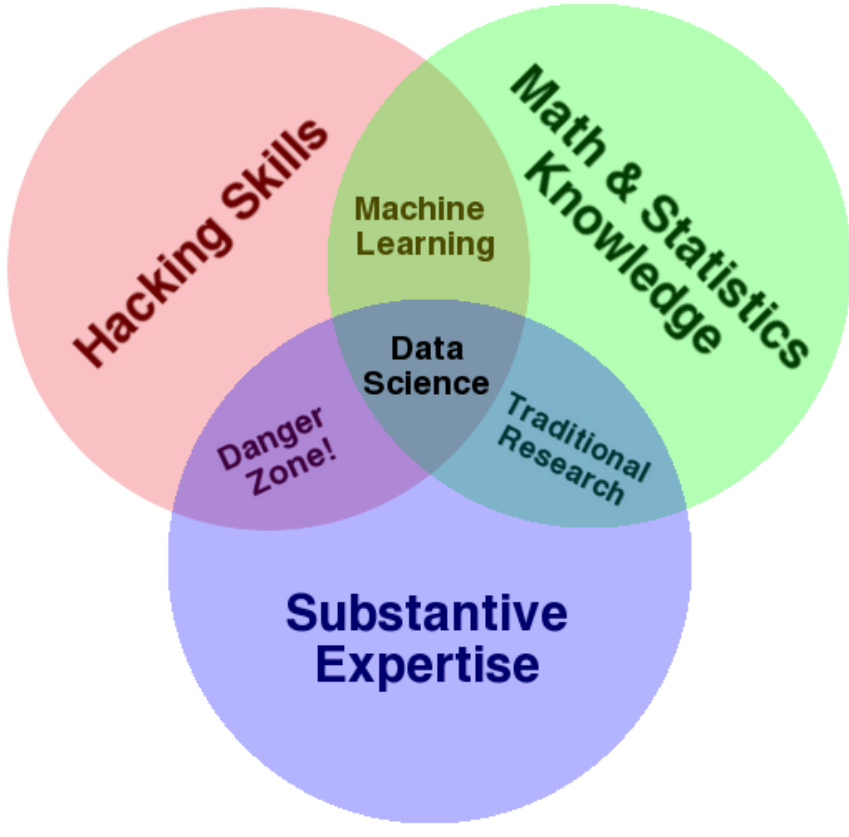
- With probability α, follow an outgoing link; otherwise jump uniformly to one of N pages.
- A dangling page (no outlinks) distributes its rank uniformly.

$$r_i = \frac{1-\alpha}{N} + \alpha \left(\sum_{j \to i} \frac{r_j}{d_j} + \frac{1}{N} \sum_{j : d_j = 0} r_j \right).$$

Here $0 < \alpha < 1$, d_j is the outdegree, and $\sum_i r_i = 1$.

Class

Data Science



Data Science Venn Diagram. By Drew Conway

Course Outline

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  Scrape it, clean it
  Basics of webscraping
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